A complete small-order domain for the subgroup-atlas reconstruction

Private L054 recovery component

Boundary and retained inputs. The fixed 2013 article, Section 4, physical pages 5–6, restricts its small-order assertion to almost simple groups appearing in its cited 2012 online ATLAS. The final 2015 version, page 264, has the same qualification. We instead construct a complete mathematical superset. This does not assert historical file identity. Nor does it establish the correctness of any unverified, named larger companion table.

We explicitly retain the classified nonabelian-simple-group roster below 10^6 documented in GAP 4.15.1, Reference Manual 39.15–15 and 39.15–17, and implemented in the pinned `grp/simple.gi` and `simple.gd`. Its non-PSL_2 data give nineteen entries, of which A_5 and A_6 are respectively $\mathrm{PSL}_2(5)$ and $\mathrm{PSL}_2(9)$. We retain simplicity, the indicated exceptional isomorphisms, and the full automorphism/outer-group classifications below. For classical groups the explicit presentations in `OuterAutoSimplePres` cite Bray–Holt–Roney-Dougal, pages 36–37; the same pinned implementation specifies the alternating and sporadic exceptions. These are mathematical classification inputs, not conclusions inferred from finite order tests. The ATLAS entries for $L_3(4)$ and $U_3(5)$ independently record the nonabelian outer groups D_{12} and S_3. No new proof of CFSG or of automorphism classification is claimed.

Lemma 1 (Preimages and isomorphism classes)**.** *Let S be a nonabelian finite simple group. The almost simple groups with socle S are exactly the preimages $\pi^{-1}(K)$, where $\pi : \mathrm{Aut}(S) \to \mathrm{Out}(S)$ and $K \leq \mathrm{Out}(S)$. Two such preimages are isomorphic if and only if their outer subgroups are conjugate in* $\mathrm{Out}(S)$.

Proof. Identify S with $\mathrm{Inn}(S)$. A group between S and $\mathrm{Aut}(S)$ containing one representative of an outer coset contains the whole coset, proving the preimage assertion. Its order is $|S||K|$. The centralizer of S in $\mathrm{Aut}(S)$ is trivial: if α commutes with every inner automorphism, then $c_{\alpha(s)} = \alpha c_s \alpha^{-1} = c_s$, so $\alpha(s) = s$ because $Z(S) = 1$. Any nontrivial normal subgroup N of a preimage meets S nontrivially, for otherwise $[N, S] \leq N \cap S = 1$. Simplicity then implies $S \leq N$. Thus S is the unique minimal normal subgroup and is characteristic. An isomorphism of two preimages restricts to $\beta \in \mathrm{Aut}(S)$; conjugation on S shows that its value on any element h is $\beta h \beta^{-1}$. Therefore the preimages are conjugate in $\mathrm{Aut}(S)$, exactly when their images are conjugate in $\mathrm{Out}(S)$. The converse is immediate. □

Lemma 2 (The linear rank-one part)**.** *There are precisely* 86 *almost simple isomorphism classes of order at most 10^6 whose socle is $\mathrm{PSL}_2(q)$. They have* 39 *distinct simple socles, taking $q = 5$ in place of the duplicate $q = 4$.*

Proof. For $q = p^f \geq 4$ put $d = (2, q-1)$. Counting ordered bases gives $|\mathrm{GL}_2(q)| = (q^2-1)(q^2-q)$. Determinant is onto $\mathbb{F}_q^\times$, and the center of $\mathrm{SL}_2(q)$ has order d. Hence $|S| = q(q^2-1)/d$. The projective action has precisely the scalar kernel: a linear map fixing every one-dimensional subspace fixes the two coordinate lines and their sum, forcing equal diagonal entries. The retained automorphism theorem is $\mathrm{Aut}(S) = \mathrm{P\Gamma L}_2(q)$, including $q = 9$. The determinant modulo squares and the field automorphism give $\mathrm{Out}(S) = C_d \times C_f$. The field action on the order-two diagonal quotient is trivial (odd p acts trivially modulo two).

For $q \geq 127$, the lower bound $q(q^2-1)/2$ exceeds 10^6 and is increasing. Thus it suffices to test prime powers $5 \leq q \leq 126$. The literal program `roster.py` tests primality by trial division, lists these prime powers, and forms every subgroup of the small explicit additive group $\mathbb{Z}/d \times \mathbb{Z}/f$. Its generic procedure starts with the identity subgroup and adjoins every possible element to each known subgroup, taking closure; every finite generating sequence is therefore represented. A separate check generates every pair of elements. This also suffices: projection of any subgroup onto C_f is cyclic, and the kernel is contained in the cyclic group C_d; a lift of a projection generator and a kernel generator generate the subgroup. The two inventories agree. Keeping precisely those K with $q(q^2-1)|K|/d \leq 10^6$ gives 86 entries. Since the outer group is abelian, no further fusion occurs. The only duplicate simple name in this range is $\mathrm{PSL}_2(4) = \mathrm{PSL}_2(5)$, removed explicitly; the other listed linear socle orders are distinct. □

Theorem 1 (Conditional classified roster)**.** *With the explicit classified inputs above, there are exactly* 127 *almost simple isomorphism classes of order at most* 10^6*. The accepted eighty current small tables cover eighty of them; the forty-seven additional targets are all in the* PSL_2 *family.*

Proof. The remaining seventeen simple socles and their retained outer groups are shown below. D_{12} denotes the dihedral group of order twelve. The last column counts conjugacy classes of outer subgroups whose preimages satisfy the bound.

S	$\lvert S\rvert$	$\mathrm{Out}(S)$	count
A_7, A_8, A_9	$2520, 20160, 181440$	C_2	$2+2+2$
$L_3(3)$	5616	C_2	2
$L_3(4)$	20160	D_{12}	10
$L_3(5)$	372000	C_2	2
$U_3(3)$	6048	C_2	2
$U_3(4)$	62400	C_4	3
$U_3(5)$	126000	S_3	4
$U_4(2) = \mathrm{PSp}_4(3)$	25920	C_2	2
$\mathrm{Sp}_4(4)$	979200	C_4	1
$\mathrm{Sz}(8)$	29120	C_3	2
M_{11}	7920	1	1
M_{12}	95040	C_2	2
M_{22}	443520	C_2	2
J_1	175560	1	1
J_2	604800	C_2	1

The program explicitly multiplies the relevant cyclic and dihedral groups, closes all subgroups and fuses by every inner conjugation. Their 41 surviving classes, together with the 86 linear classes, give 127. The checked roster assigns all eighty accepted targets once, with exact outer-index multiplicities. Their identities as the stated almost simple constructions are acquired from their accepted named-model transports, not deduced from orders. Where several outer classes have the same order, distinctness is additionally forced by the accepted numbers of subgroup conjugacy classes: $25, 26, 56$ for the three $A_6.2$ groups; $43, 103, 39$ for the three $L_2(25).2$ groups; $151, 152, 102$ for the three $L_3(4).2$ groups; and $143, 226, 142$ for the three index-six $L_3(4)$ extensions. These are intrinsic isomorphism invariants. Each relevant multiplicity is therefore exhausted, with no assumption based on a displayed name. The difference is $127 - 80 = 47$. □

Actual finite applications and limits. The forty-seven additions comprise twenty prime-field parent descents, eight nonprime parent descents and nineteen index-two restrictions. Each uses the previously acquired complete-local-cover descent or normal-restriction theorem; source counts are never inputs. The projective models of orders 235200, 531360, 885720, 976500, 524160 and 786240 are literal defining actions, with full generator and receipt records. Final

states use the accepted explicit generator serializer. The separate two-page applications proof and exact literal certificates now cover every new row. This roster argument supplies the domain classification; distinct acceptance of the combined component remains a separate review step.

Section 4 does not name an individual larger group: it refers to some larger groups, including large sporadic groups, and then reports an atlas of more than a hundred outputs. The actual companion index supplies individual names. A fully verified 127-table reconstruction would prove the complete small-order superset and a stronger cardinality statement. It would not by itself verify every larger PDF named in that index or prove the exact historical output set. Those claims must retain their own source-qualified dispositions.